

Week 4 — Pipeline Creation & Model Investigation

Deliverable 4: End-to-End KNIME Prediction Pipeline. Tasks:

- **Pipeline Construction:** Build a workflow:
CSV Reader → Missing Value → Row Filter → Math Formula → One to Many → Normalizer → Partitioning → Linear Regression Learner → Regression Predictor → Numeric Scorer

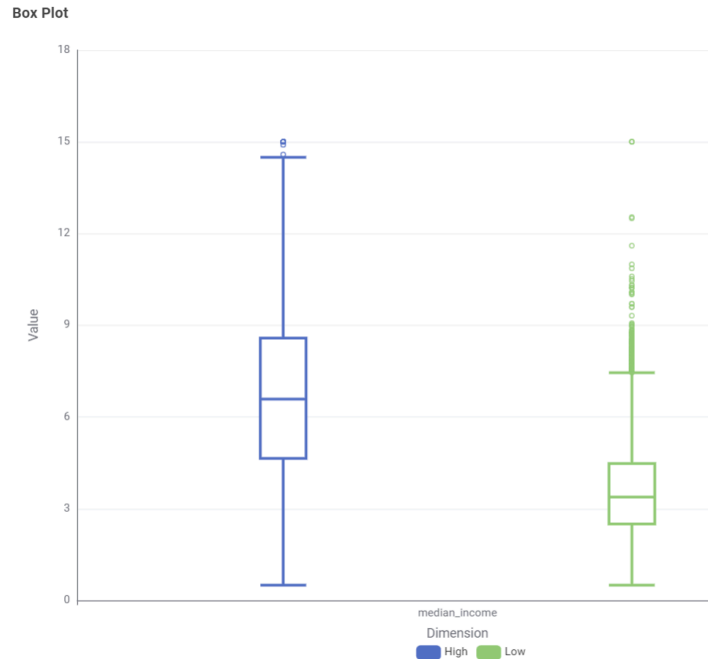
☐	#	RowID	Prediction (median_house_value) .00 Number (Float)
☐	1	R^2	0.638
☐	2	mean absolute error	51,239.438
☐	3	mean squared error	4,930,397,519.496
☐	4	root mean squared error	70,216.789
☐	5	mean signed difference	1,046.163
☐	6	mean absolute percentage error	0.293
☐	7	adjusted R^2	0.638

- **Bias Check:** Your model overpredicts houses above \$400k. Use **GroupBy** and **Box Plots** to explain why.
Nodes: Math Formula, GroupBy, Box Plot, Annotation.
- **Target Swap:** Build a sub-model to predict **median_income** instead of house value to identify feature importance. Nodes: Column Filter, Linear Regression Learner, Regression Predictor, Numeric Scorer, Bar Chart, Annotation.

Rows: 170 | Columns: 2

Table  Statistics 

☐	#	RowID	median_house_value .00 Number (Float)	Prediction (median_house_value) .00 Number (Float)
☐	1	Row1	358,500	449,049.291
☐	2	Row12	391,100	400,609.381
☐	3	Row12	295,200	418,635.076
☐	4	Row15	348,700	411,104.045
☐	5	Row40	430,500	457,183.877
☐	6	Row51	500,001	617,088.247
☐	7	Row51	500,001	583,817.254
☐	8	Row51	500,001	450,154.161
☐	9	Row57	350,700	421,034.967
☐	10	Row92	500,001	419,328
☐	11	Row97	475,800	419,622.126
☐	12	Row15	500,001	493,756.164
☐	13	Row15	500,001	423,936
☐	14	Row15	433,300	406,409.862
☐	15	Row16	500,001	472,372.147



Rows: 2 | Columns: 5

Table Statistics

#	RowID	price_category T String	Mean(median_income) Number (Float)	Mean(rooms_per_house... Number (Float)	Mean(population) Number (Float)	Mean(median_house_value) Number (Float)
1	Row0	High	6.931	6.501	1,172.411	475,000.553
2	Row1	Low	3.588	5.33	1,448.833	182,107.488

The model tends to overpredict homes above \$400,000 because high-value homes are strongly associated with higher median income and distinct feature patterns. These homes are less frequent in the dataset, causing the model to generalize poorly and produce biased predictions. The target swap model shows that features such as location and housing characteristics significantly influence median income. Since income is strongly correlated with house value, this explains why the original model relied heavily on income-related patterns when predicting housing prices.



Team Weekly Sync: Project Checklist

Group Name/Number: Group 5

| **Date:** 4/12/2026

Use this checklist at the start of every meeting to ensure you are meeting the "Green-to-Gold" requirements and project milestones.

1. Infrastructure & Accountability (Start-of-Meeting)

- [X] **GitHub Sync:** Did everyone pull the latest changes from `main`?
- [X] **Contribution Log:** Has the `README.md` been updated to reflect progress made this week?
- [X] **Blocking Issues:** Is anyone currently "blocked" by a KNIME node error or a data issue? *(If yes, resolve this first!)*

2. Current Sprint Focus (Check the Week)

- **Week 2 (EDA):** Are your distributions analyzed?
- **Week 3 (Prep):** Is your feature matrix complete?
- **Week 4 (Modeling):** Is your data analysis or predictive pipeline running end-to-end?
- **Week 5 (Fairness):** Have you compared at least two corrective strategies (e.g., threshold adjustment vs. oversampling)?
- **Week 6 (Presentation):** Does your workflow run "Green-to-Gold" on new data?

3. Technical Quality Assurance (The "Green-to-Gold" Audit)

- [X] **Automation:** Does the workflow run from `File Reader` to `Output` without changing settings?
- [X] **Annotation:** Are your primary logic steps (e.g., normalisation, optimization) explained in a **Metanode Annotation**?
- [X] **Evidence:** Is every chart/table in your report directly linked to a specific KNIME node output?

4. Action Items (End-of-Meeting)

Who is doing what before next week?

Task	Assignee	Deadline
Address overprediction question	Brian	4/17/2026
	Rachel	4/17/2026
	Isaiah	4/17/2026